science. Kuhn argued that both the observable and unobservable content of scientific worldviews—what he called “paradigms”—play vital roles in the actual practice of science. These scientific paradigms influence what experiments are done, how they’re performed, and how the results are interpreted. Going back to the example of the malfunctioning remote control, replacing the batteries is a reasonable thing to do because your knowledge of remote controls, televisions, and batteries suggests that dead batteries are the most likely reason for the remote to stop working. That same set of knowledge—your “paradigm of home entertainment systems”—also tells you that it’s impossible for your TV to suddenly display all pictures in infrared and emit all sound in ultrasonic. Kuhn argued that paradigms guide scientific practice in a similar way. For example, chemists in the nineteenth century believed in atomic theory, which dictated that there was a limited number of elements, each composed of identical atoms, and those atoms combined to form compounds that had fixed ratios of each element. These ideas were central to the practice of chemistry at the time and, according to Kuhn, had the power of “setting the problem of atomic weights, bounding the admissible results of chemical analyses, and informing chemists what atoms and molecules, compounds and mixtures, were.” At every step of the way—forming hypotheses, designing and conducting experiments, even simply observing the results of those experiments—the paradigm of atomic theory informed the actions of the nineteenth-century chemists. And they were wildly successful, discovering the periodic table of the elements decades before physicists discovered electrons or learned anything about atomic structure. Yet, according to the best science of the time, atoms were unobservable. So, Kuhn concluded, it’s not just the observable part of a theory that matters—the full content of scientific paradigms influences how science is done. The interpretation of a physical theory, like quantum physics, is important for the day-to-day practice of science itself. Logical positivism couldn’t account for this.

What, precisely, Kuhn advocated in place of positivism is not entirely clear. And some of his bolder claims, about the inability to rationally compare competing scientific theories, were dismissed as mistakes and didn’t catch on with professional philosophers of science. But Kuhn’s criticism of positivism and his observations on scientific practice were widely seen as accurate—and he wasn’t the only one pointing these things